

Comparative CV / ML / DL Models and a Final Hybrid Architecture

*Research planning document for the collaborative brain-tumor detection, classification,
and segmentation project*

Central question

How can we predict the presence of brain tumors using a hybrid approach of computer vision, machine learning, and deep learning?

Comparative question

Which model provides the best performance for tumor classification, detection, and/or segmentation under controlled experimental conditions?

End goal

Evaluate eight complementary models, then construct a ninth hybrid model using the strongest complementary features and representations discovered experimentally.

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Working research reference - Faraz & Isaiah

1. Research Strategy

The project is designed as a comparative study spanning classical computer vision (CV), conventional machine learning (ML), deep learning (DL), image segmentation, object detection, and feature-level fusion. The eight initial models are intended to answer different technical questions rather than represent minor variations of the same neural network. Their results then provide evidence for selecting components of a final hybrid model.

A key methodological distinction is that classification, detection, and segmentation are related but different tasks. Classification asks whether a tumor is present or what type it is; object detection additionally localizes the lesion with a bounding box; segmentation delineates the tumor at the pixel or voxel level. The project can incorporate all three, but their metrics should not be conflated.

2. Eight-Model Experimental Framework

Model	Architecture	Family	Primary role
1	HOG + LBP -> SVM	CV + ML	Handcrafted gradient/texture baseline
2	GLCM + Wavelets -> Random Forest	CV + ML	Texture/statistical baseline
3	Combined handcrafted features -> XGBoost	CV + ML	Strong nonlinear boosted-tree baseline
4	Custom CNN	DL	End-to-end learned-feature baseline
5	EfficientNet transfer learning	DL	Pretrained deep classifier
6	U-Net / Attention U-Net	DL + segmentation	Pixel/voxel-level tumor delineation
7	CNN deep features -> SVM/XGBoost	Hybrid DL + ML	Learned representations with classical classifier
8	YOLO tumor detector	CV + DL	Bounding-box localization and detection
9	Final feature-fusion hybrid	CV + ML + DL	Fuse strongest complementary components from Models 1-8

Model 1 - HOG + LBP -> SVM

Histogram of Oriented Gradients + Local Binary Pattern -> Support Vector Machine

Category: Computer Vision + Machine Learning

HOG describes local shape and edge structure through gradient orientations. LBP captures local texture by comparing neighboring pixel intensities. SVM learns a separating decision boundary in the resulting feature space. PCA (Principal Component Analysis) can optionally reduce dimensionality before classification.

Proposed pipeline: MRI -> preprocessing -> HOG/LBP feature extraction -> scaling/PCA -> SVM -> tumor class.

Enhancing multiclass brain tumor diagnosis using SVM and innovative feature extraction techniques (2024)

Directly relevant to the proposed baseline: HOG, LBP, PCA, and SVM are used for multiclass brain-tumor MRI diagnosis.

[Open publication / source](#)

Brain tumor magnetic resonance images classification based machine learning paradigms (2023)

Compares handcrafted representations including GLCM, HOG, and LBP with multiple machine-learning classifiers, including SVM and KNN.

[Open publication / source](#)

Model 2 - GLCM + Wavelets -> Random Forest

Gray-Level Co-occurrence Matrix + Wavelet Features -> Random Forest

Category: Computer Vision + Machine Learning

GLCM quantifies spatial relationships among gray levels and supports texture descriptors such as contrast, correlation, energy, homogeneity, and dissimilarity. Wavelet transforms decompose an image across spatial scales/frequencies. Random Forest combines many decision trees into an ensemble classifier.

Proposed pipeline: MRI -> GLCM/wavelet/statistical texture features -> Random Forest -> tumor class.

Brain tumor classification from multi-modality MRI using wavelets and machine learning (2017)

An older but methodologically close precedent using intensity, neighborhood, and wavelet texture information with Random Forest classification.

[Open publication / source](#)

Feature extraction from MRI ADC images for brain tumor classification using machine learning techniques (2022)

Uses GLCM and statistical features such as entropy, energy, contrast, homogeneity, correlation, variance, and kurtosis; Random Forest is among the evaluated classifiers. ADC stands for Apparent Diffusion Coefficient.

[Open publication / source](#)

Comparative Analysis of MRI Image Classification Methods Based on Texture Features for Brain Tumor Detection Using Random Forest (2026)

Recent study emphasizing histogram, GLCM, and GLRLM (Gray-Level Run-Length Matrix) texture features with Random Forest.

[Open publication / source](#)

Model 3 - Combined CV Features -> XGBoost

Combined Computer-Vision Features -> Extreme Gradient Boosting

Category: Computer Vision + Machine Learning

This model concatenates complementary handcrafted descriptors - potentially HOG, LBP, GLCM, wavelets, intensity statistics, and shape features - then trains XGBoost. XGBoost (Extreme Gradient Boosting) builds decision trees sequentially so later trees correct residual errors from earlier trees.

Proposed pipeline: MRI -> HOG + LBP + GLCM + wavelets + intensity/shape statistics -> feature selection/PCA -> XGBoost.

Brain tumor magnetic resonance images classification based machine learning paradigms (2023)

Provides direct evidence for the usefulness of HOG/LBP/GLCM handcrafted representations and a comparative ML framework that can be extended to XGBoost.

[Open publication / source](#)

Feature extraction from MRI ADC images for brain tumor classification using machine learning techniques (2022)

Supports a statistical-feature extraction and feature-selection workflow that can be paired with boosted-tree classification.

[Open publication / source](#)

Research opportunity: the exact HOG + LBP + GLCM + wavelet -> XGBoost configuration appears less standardized than the SVM and Random Forest baselines, leaving room for an original controlled comparison.

Model 4 - Custom CNN

Custom Convolutional Neural Network

Category: Deep Learning

A CNN learns hierarchical image features directly from MRI pixels. A compact baseline might use repeated Convolution -> Batch Normalization -> ReLU (Rectified Linear Unit) -> Pooling blocks, followed by Global Average Pooling and a classification head. This tests whether learned features outperform handcrafted CV descriptors.

Enhancing brain tumor classification in MRI scans with a multi-layer customized convolutional neural network approach (2024)

Uses a custom CNN on thousands of MRI images representing glioma, meningioma, pituitary tumor, and no-tumor categories.

[Open publication / source](#)

Brain Tumor Classification in MRI Images Using Combined Transfer Learning and Convolutional Neural Networks (2026)

Compares a custom CNN with multiple transfer-learning architectures, providing a useful precedent for the project's controlled model-comparison philosophy.

[Open publication / source](#)

Model 5 - EfficientNet Transfer Learning

EfficientNet Convolutional Neural Network with Transfer Learning

Category: Deep Learning / Transfer Learning

Transfer learning adapts a network pretrained on a large source dataset to a new target task. EfficientNet jointly scales network depth, width, and input resolution. A practical workflow is: pretrained EfficientNet -> replace classification head -> train head on MRI -> fine-tune selected backbone layers.

Multi-class classification of brain tumor types from MR images using EfficientNets (2023)

Fine-tunes EfficientNet variants for glioma, meningioma, and pituitary tumor classification and uses Grad-CAM (Gradient-weighted Class Activation Mapping) for explainability.

[Open publication / source](#)

Multigrade brain tumor classification in MRI images using Fine tuned EfficientNet (2024)

Uses fine-tuned EfficientNetV2S and Grad-CAM for MRI brain-tumor classification.

[Open publication / source](#)

Brain Tumor Classification and Segmentation in MR Images Using EfficientNet and U-Net++ Models (2026)

Especially relevant to the eventual hybrid system because EfficientNet performs classification while U-Net++ performs segmentation in the same overall study.

[Open publication / source](#)

Model 6 - U-Net / Attention U-Net

U-Shaped Convolutional Neural Network / Attention U-Net

Category: Deep Learning + Image Segmentation

U-Net predicts a dense segmentation mask rather than only an image-level class. Attention variants introduce mechanisms that help emphasize relevant tumor regions. For BraTS-style evaluation, use segmentation metrics such as DSC (Dice Similarity Coefficient), IoU (Intersection over Union), and HD95 (95th-percentile Hausdorff Distance).

AttU-NET: attention U-net for brain tumor segmentation (2022)

Applies an attention-enhanced U-Net to brain-tumor segmentation using BraTS (Brain Tumor Segmentation Challenge) data.

[Open publication / source](#)

MRI Brain Tumour Segmentation Using Multiscale Attention U-Net (2024)

Evaluates a multiscale attention U-Net on BraTS 2019, 2020, and 2021 datasets and reports region-specific segmentation performance.

[Open publication / source](#)

Self-attention U-Net (SAU-Net): An attention-driven U-Net framework for precise brain tumor segmentation using multimodal magnetic resonance imaging (2026)

Uses self-attention within a U-Net-style framework and evaluates multimodal MRI brain-tumor segmentation on BraTS datasets.

[Open publication / source](#)

Model 7 - CNN Deep Features -> SVM / XGBoost

Convolutional Neural Network Feature Extraction -> Support Vector Machine / Extreme Gradient Boosting

Category: Hybrid Deep Learning + Machine Learning

Here the CNN is used as a representation learner rather than the final classifier. A late-layer deep feature vector is extracted and passed to SVM or XGBoost. This directly tests whether learned deep representations and classical ML classification complement one another.

MRI-Based Brain Tumor Classification Using Ensemble of Deep Features and Machine Learning Classifiers (2021)

Uses pretrained CNNs as deep feature extractors, fuses deep representations, and evaluates classical ML classifiers; RBF-SVM (Radial Basis Function Support Vector Machine) is particularly relevant.

[Open publication / source](#)

Deep Feature Extraction with SVM Classifier for Brain Tumor Classification (2024)

Uses CLAHE (Contrast Limited Adaptive Histogram Equalization), deep CNN feature extractors such as VGG/ResNet, and SVM classification.

[Open publication / source](#)

Brain tumor classification from MRI images using a multi-scale channel attention CNN integrated with SVM (2026)

A recent hybrid CNN + SVM approach combining multiscale learned features, channel attention, and classical classification.

[Open publication / source](#)

Model 8 - YOLO Brain Tumor Detector

You Only Look Once Object Detection

Category: Computer Vision + Deep Learning + Object Detection

YOLO (You Only Look Once) is a family of single-stage object detectors. Unlike image-level classification, object detection predicts both a tumor class/confidence and a bounding box. This model therefore bridges classification and spatial localization.

Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging (2023)

Modified YOLOv7 brain-tumor detector incorporating attention and multiscale feature components such as CBAM, SPPF, BiFPN, and PANet.

[Open publication / source](#)

Application of algorithms based on improved YOLO in MRI image detection of brain tumors (2025)

Uses an improved YOLO11n design with components including GhostConv, OREPA, and efficient multi-scale attention.

[Open publication / source](#)

Brain tumor detection in MRI using YOLO architectures (2026)

Recent comparative work examining multiple YOLO architectures for MRI brain-tumor detection and localization.

[Open publication / source](#)

3. Model 9 - Final Hybrid Feature-Fusion Model

Model 9 should be constructed only after Models 1-8 have been evaluated. The project should not assume in advance that every technique belongs in the final architecture. Instead, the empirical results should identify complementary representations worth fusing.

Illustrative architecture:

MRI input

- |-- Classical CV branch: HOG / LBP / GLCM / wavelets -> handcrafted feature vector
- |-- Deep branch: EfficientNet (or best CNN) -> deep feature vector
- |-- Segmentation branch: U-Net / Attention U-Net -> ROI and morphology features
- `-- Optional detection branch: YOLO -> localization/confidence features
 - > feature normalization / selection / fusion
 - > final SVM, XGBoost, or compact neural classifier
 - > calibrated tumor probability + optional uncertainty estimate

Feature-Level Fusion

A practical representation can concatenate three principal feature families:

- F_CV: handcrafted computer-vision features such as HOG, LBP, GLCM, wavelets, and intensity statistics.
- F_DL: deep features extracted from a CNN or EfficientNet layer before the final classifier.
- F_SEG: segmentation-derived features such as tumor area/volume, perimeter/surface, compactness, location, and other ROI descriptors.

The combined feature vector $F_{\text{Hybrid}} = [F_{\text{CV}}, F_{\text{DL}}, F_{\text{SEG}}]$ can then be evaluated with SVM, XGBoost, or a compact neural classifier. Feature selection and dimensionality reduction should be performed within the training folds to avoid information leakage.

4. Evaluation and Experimental Controls

All models should be compared under a common protocol whenever the task permits. Patient-level data splitting is essential when multiple images/slices come from the same subject, because image-level random splitting can cause train-test leakage and inflate performance.

Classification Metrics

- Accuracy
- Precision / Positive Predictive Value
- Recall / Sensitivity
- Specificity
- F1 score
- ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)
- Confusion matrix
- Calibration metrics and confidence intervals where appropriate

Segmentation Metrics

- DSC - Dice Similarity Coefficient
- IoU - Intersection over Union (Jaccard Index)
- HD95 - 95th-percentile Hausdorff Distance
- Sensitivity / recall
- Precision

Detection Metrics

- mAP - mean Average Precision
- IoU-based localization quality
- Precision and recall
- Per-class Average Precision where multiple tumor types are detected

Controlled Experimental Factors

- Same patient-level train/validation/test partitions for comparable models
- Same image resizing and normalization where appropriate
- Documented preprocessing and augmentation
- Fixed/random seeds reported
- Class imbalance strategy documented
- Hyperparameter tuning restricted to training/validation data
- Final test set used only after model selection
- Statistical comparison or confidence intervals for important performance differences

5. Abbreviation and Terminology Reference

Abbreviation	Full term	Meaning / role
AI	Artificial Intelligence	Broad field of computational systems performing tasks associated with intelligence
ML	Machine Learning	Algorithms that learn predictive patterns from data
DL	Deep Learning	Machine learning based on multilayer neural networks
CV	Computer Vision	Computational analysis and interpretation of images
MRI	Magnetic Resonance Imaging	Medical imaging modality
CNN	Convolutional Neural Network	Neural network architecture specialized for spatial/image data
HOG	Histogram of Oriented Gradients	Gradient/shape feature descriptor
LBP	Local Binary Pattern	Local texture descriptor
GLCM	Gray-Level Co-occurrence Matrix	Statistical texture representation
GLRLM	Gray-Level Run-Length Matrix	Run-length texture representation
SVM	Support Vector Machine	Supervised classifier based on separating decision boundaries
RBF	Radial Basis Function	Common nonlinear kernel for SVM
RF	Random Forest	Ensemble of decision trees
KNN	K-Nearest Neighbors	Instance-based classifier
PCA	Principal Component Analysis	Linear dimensionality-reduction method
XGBoost	Extreme Gradient Boosting	Gradient-boosted decision-tree implementation

ReLU	Rectified Linear Unit	Neural-network activation function
ResNet	Residual Network	CNN family using residual/skip connections
VGG	Visual Geometry Group	Oxford group/name associated with VGG CNN architectures
CLAHE	Contrast Limited Adaptive Histogram Equalization	Local contrast-enhancement technique
XAI	Explainable Artificial Intelligence	Methods for interpreting model behavior
Grad-CAM	Gradient-weighted Class Activation Mapping	CNN explanation/visualization technique
YOLO	You Only Look Once	Single-stage object-detection family
CBAM	Convolutional Block Attention Module	Channel/spatial attention mechanism
SPPF	Spatial Pyramid Pooling Fast	Multi-scale feature aggregation module
BiFPN	Bidirectional Feature Pyramid Network	Bidirectional multi-scale feature fusion
PANet	Path Aggregation Network	Feature aggregation architecture
ROI	Region of Interest	Target image region, such as a tumor
BraTS	Brain Tumor Segmentation Challenge	Benchmark/challenge for brain-tumor MRI segmentation
DSC	Dice Similarity Coefficient	Segmentation overlap metric
IoU	Intersection over Union	Overlap metric for segmentation/detection
HD95	95th-percentile Hausdorff Distance	Robust segmentation-boundary distance metric
ADC	Apparent Diffusion Coefficient	Quantitative diffusion MRI measurement
ROC	Receiver Operating Characteristic	Curve of true-positive vs false-positive rate
AUC	Area Under the Curve	Summary of ROC discrimination performance
mAP	mean Average Precision	Common object-detection performance metric

6. Recommended Implementation Sequence

1. Lock the dataset(s), patient-level splits, preprocessing, image size, augmentation policy, and evaluation metrics.
2. Implement Models 1-3 as classical CV/ML baselines.
3. Implement Model 4 as a transparent custom-CNN baseline.
4. Implement Model 5 using EfficientNet transfer learning.
5. Implement Model 6 on data with segmentation masks; compare U-Net and an attention variant if resources permit.
6. Implement Model 7 by extracting deep features from the best CNN/EfficientNet and training SVM/XGBoost on those representations.
7. Implement Model 8 on data with bounding boxes or derive carefully validated boxes from masks where scientifically appropriate.
8. Perform controlled comparison, error analysis, and statistical uncertainty estimation.
9. Construct Model 9 only from components shown to provide complementary value, then compare it against the strongest single-model baseline.

7. Compact Literature Matrix

Model	Method	Paper / short title	Year	Relevance
1	HOG/LBP + SVM	Basthikodi et al.	2024	HOG/LBP/PCA/SVM multiclass MRI diagnosis
1	CV/ML comparison	Brain tumor MRI classification based ML paradigms	2023	GLCM/HOG/LBP with multiple classifiers
2	Texture + RF	Feature extraction from MRI ADC images	2022	GLCM/statistical features + ML
2	Texture + RF	Comparative MRI texture classification with RF	2026	GLCM/GLRLM + Random Forest
4	Custom CNN	Customized multi-layer CNN	2024	End-to-end CNN classification
4	CNN comparison	Abbas et al.	2026	Custom CNN vs transfer-learning models
5	EfficientNet	EfficientNets for multi-class brain tumors	2023	EfficientNet variants + Grad-CAM
5	EfficientNet	Fine-tuned EfficientNet	2024	EfficientNetV2S + Grad-CAM
5/6	Classification + segmentation	EfficientNet + U-Net++	2026	Combined classification/segmentation workflow
6	Attention U-Net	AttU-NET	2022	Attention U-Net segmentation
6	Attention U-Net	Multiscale Attention U-Net	2024	BraTS segmentation
6	Self-attention U-Net	SAU-Net	2026	Multimodal MRI segmentation
7	Deep features + ML	Ensemble deep features + ML classifiers	2021	CNN feature fusion + classical ML
7	Deep features + SVM	Deep Feature Extraction with SVM	2024	CLAHE + CNN features + SVM
7	CNN + SVM	Multi-scale channel attention CNN + SVM	2026	Hybrid learned features + SVM
8	YOLO	Abdusalomov et al.	2023	Modified YOLOv7 tumor detection
8	YOLO	Improved YOLO MRI detection	2025	YOLO11n improvements
8	YOLO comparison	Hammad & Makram	2026	Comparative YOLO architectures

8. Research Notes and Scope

This document is a research-planning synthesis, not a finalized protocol or systematic review. Publication links are included as starting points for full-text review. Before citing numerical performance in a manuscript, verify the exact dataset, split strategy, sample size, class balance, preprocessing, and metric definition in the original paper. Reported accuracies across different papers are not directly comparable unless experimental conditions are comparable.

The final manuscript should clearly distinguish source-derived claims from new experimental results produced by this collaboration. The strongest contribution may be the controlled comparison across classical CV, ML, DL, segmentation, detection, and hybrid feature-fusion approaches under a common evaluation framework.